He ZiAng

System Engineer | Manufacturing Software & Data Integration

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Professional summary

System Engineer with experience since 2022 developing and supporting manufacturing applications, data pipelines, and equipment monitoring systems. Work spans React interfaces, backend APIs, Python data processing, Oracle and PostgreSQL integration, and quality inspection workflows.

Work experience

Resonac HD Singapore 2022.03 - Present
System Engineer Singapore

- Developed a visual inspection application for defect tagging, inspection judgements, configurable rules, audit history, and shift reports using React/TypeScript and Bun/Express. Prepared Docker-based UAT deployment with Oracle and inspection-image integration.
- Built the Guzik measurement-data platform with Python/Oracle ingestion and Excel reporting, plus React/Express applications for engineering test-plan configuration, target management, regeneration, and measurement review.
- Extended the OMS measurement-management system using ASP.NET/VB.NET, with instrument configuration, requests, acknowledgements, result uploads, and statistical process control (SPC) integration. Implemented pKerr instrument workflows and contributed to FTIR reporting.
- Developed Python equipment-log pipelines with processing configurations for 11 production lines, converting sputtering logs into Parquet datasets, extracting process cycles, tracking Oracle metadata, and archiving data. Built Flask visualization applications and added connection pooling and processing diagnostics.
- Developed and supported pump, motor-torque, and process-parameter monitoring using Python, PostgreSQL, MQTT, and web APIs. Implemented threshold alerts, PLC-connected cassette tracking, WebSocket broadcasts, and exception-log ingestion.
- Contributed to a shared Workplace-to-Teams migration documented as completed for 9 manufacturing alert integrations. Updated message formatting, caller configuration, and application integrations through a shared Microsoft Graph service.
- Enhanced a shared React job-request application and its API integration with approval routing, division-manager auto-approval, and email notifications.
- Used LLM coding assistants, including Codex and Claude Code, for development, refactoring, debugging, and test preparation, with review and verification of generated changes.

Vanitee 2017.09 - 2018.02
iOS Developer Intern Singapore

- Worked on a Swift business application for iPhone and iPad. The project was discontinued when the business moved to a web-based system.
- Integrated React Native into the iOS application to support code sharing with Android.

Technical skills

Languages: Python, JavaScript, TypeScript, SQL, VB.NET; earlier experience with Swift and Java.

Applications and APIs: React, Express, Bun, Flask, ASP.NET, REST APIs.

Data and integration: Oracle, PostgreSQL, pandas, Parquet, MQTT, WebSockets, SPC integration.

Delivery and support: Git, Docker, UAT preparation, database connection pooling, application logging, and troubleshooting.

Education

University of New South Wales 2018 - 2020
Computer Science Bachelor's degree

Ngee Ann Polytechnic 2015 - 2018
Information Technology Diploma

Languages

Chinese [Native] / English [Professional]